THE ECONOMIST

LEADERS • ARTIFICIAL INTELLIGENCE & SCIENTIFIC PROGRESS

# How AI can revolutionise science

The technology is being applied in many fields—and could lead to a surge in scientific progress

## VOCABULARY

words are quoted from the article

| TERM & IPA | POS | DEFINITION | IN THE ARTICLE | 中文（先自己写） |
|---|---|---|---|---|
| **scrutiny /ˈskruːtəni/** | n. | careful and thorough examination | "Such claims are worth examining" | |
| **mechanism /ˈmekənɪzəm/** | n. | the way in which something works or produces a result | "the mechanism by which AI will supposedly solve" | |
| **simulation /ˌsɪmjuˈleɪʃn/** | n. | a model that imitates a real process or system | "based on simulation and modelling" | |
| **interdisciplinary /ˌɪntədɪsəˈplɪnəri/** | adj. | involving two or more academic subjects | "stimulate interdisciplinary work" | |
| **feasible /ˈfiːzəbl/** | adj. | possible to do successfully | "is therefore feasible" | |
| **replicate /ˈreplɪkeɪt/** | v. | to repeat something so as to obtain the same result | "easy to replicate" | |
| **adoption /əˈdɒpʃn/** | n. | the act of starting to use a method or idea | "the degree of adoption varies widely" | |
| **foster /ˈfɒstə(r)/** | v. | to encourage the development of something | "foster innovation at the boundaries between fields" | |
| **advent /ˈædvent/** | n. | the arrival or beginning of something important | "described the advent of new scientific instruments" | |
| **algorithmic /ˌælgəˈrɪðmɪk/** | adj. | relating to a set of rules followed by a computer | "algorithmic bias and discrimination" | |

## PRE-READING INQUIRY & DISCUSSION

1. The article compares AI with earlier tools such as the microscope and the computer. Which earlier change do you think is the closest parallel, and why?
2. The author says the main barrier is "sociological" rather than technical. What would that look like in your own school or workplace?

DEBATE about artificial intelligence (AI) tends to focus on its potential dangers: algorithmic bias and discrimination, the mass destruction of jobs and even, some say, the extinction of humanity. As some observers fret about these dystopian scenarios, however, others are focusing on the potential rewards. AI could, they claim, help humanity solve some of its biggest and thorniest problems. And, they say, AI will do this in a very specific way: by radically accelerating the pace of scientific discovery, especially in areas such as medicine, climate science and green technology. Luminaries in the field such as Demis Hassabis and Yann LeCun believe that AI can turbocharge scientific progress and lead to a golden age of discovery. Could they be right?

Such claims are worth examining, and may provide a useful counterbalance to fears about large-scale unemployment and killer robots. Many previous technologies have, of course, been falsely hailed as panaceas. The electric telegraph was lauded in the 1850s as a herald of world peace, as were aircraft in the 1900s; pundits in the 1990s said the internet would reduce inequality and eradicate nationalism. But the mechanism by which AI will supposedly solve the world's problems has a stronger historical basis, because there have been several periods in history when new approaches and new tools did indeed help bring about bursts of world-changing scientific discovery and innovation.

In the 17th century microscopes and telescopes opened up new vistas of discovery and encouraged researchers to favour their own observations over the received wisdom of antiquity, while the introduction of scientific journals gave them new ways to share and publicise their findings. The result was rapid progress in astronomy, physics and other fields, and new inventions from the pendulum clock to the steam engine—the prime mover of the Industrial Revolution.

Then, starting in the late 19th century, the establishment of research laboratories, which brought together ideas, people and materials on an industrial scale, gave rise to further innovations such as artificial fertiliser, pharmaceuticals and the transistor, the building block of the computer. From the mid-20th century, computers in turn enabled new forms of science based on simulation and modelling, from the design of weapons and aircraft to more accurate weather forecasting.

And the computer revolution may not be finished yet. As we report in a special Science section, AI tools and techniques are now being applied in almost every field of science, though the degree of adoption varies widely: 7.2% of physics and astronomy papers published in 2022 involved AI, for example, compared with 1.4% in veterinary science. AI is being employed in many ways. It can identify promising candidates for analysis, such as molecules with particular properties in drug discovery, or materials with the characteristics needed in batteries or solar cells. It can sift through piles of data such as those produced by particle colliders or robotic telescopes, looking for patterns. And AI can model and analyse even more complex systems, such as the folding of proteins and the formation of galaxies. AI tools have been used to identify new antibiotics, reveal the Higgs boson and spot regional accents in wolves, among other things.

All this is to be welcomed. But the journal and the laboratory went further still: they altered scientific practice itself and unlocked more powerful means of making discoveries, by allowing people and ideas to mingle in new ways and on a larger scale. AI, too, has the potential to set off such a transformation.

Two areas in particular look promising. The first is "literature-based discovery" (LBD), which involves analysing existing scientific literature, using ChatGPT-style language analysis, to look for new hypotheses, connections or ideas that humans may have missed. LBD is showing promise in identifying new experiments to try—and even suggesting potential research collaborators. This could stimulate interdisciplinary work and foster innovation at the boundaries between fields. LBD systems can also identify "blind spots" in a given field, and even predict future discoveries and who will make them.

The second area is "robot scientists", also known as "self-driving labs". These are robotic systems that use AI to form new hypotheses, based on analysis of existing data and literature, and then test those hypotheses by performing hundreds or thousands of experiments, in fields including systems biology and materials science. Unlike human scientists, robots are less attached to previous results, less driven by bias—and, crucially, easy to replicate. They could scale up experimental research, develop unexpected theories and explore avenues that human investigators might not have considered.

The idea that AI might transform scientific practice is therefore feasible. But the main barrier is sociological: it can happen only if human scientists are willing and able to use such tools. Many lack skills and training; some worry about being put out of a job. Fortunately, there are hopeful signs. AI tools are now moving from being pushed by AI researchers to being embraced by specialists in other fields.

Governments and funding bodies could help by pressing for greater use of common standards to allow AI systems to exchange and interpret laboratory results and other data. They could also fund more research into the integration of AI smarts with laboratory robotics, and into forms of AI beyond those being pursued in the private sector, which has bet nearly all its

chips on language-based systems like ChatGPT. Less fashionable forms of AI, such as model-based machine learning, may be better suited to scientific tasks such as forming hypotheses.

In 1665, during a period of rapid scientific progress, Robert Hooke, an English polymath, described the advent of new scientific instruments such as the microscope and telescope as "the adding of artificial organs to the natural". They let researchers explore previously inaccessible realms and discover things in new ways, "with prodigious benefit to all sorts of useful knowledge". For Hooke's modern-day successors, the adding of artificial intelligence to the scientific toolkit is poised to do the same in the coming years—with similarly world-changing results.

# EXERCISES

先作答，答案在最后一页

## PART 1 词汇

根据释义写出文中出现的单词

Write the word from the article that matches each definition.

1. careful and thorough examination
2. the way in which something works or produces a result
3. a model that imitates a real process or system
4. involving two or more academic subjects
5. possible to do successfully
6. to repeat something so as to obtain the same result
7. the act of starting to use a method or idea
8. to encourage the development of something
9. the arrival or beginning of something important
10. relating to a set of rules followed by a computer

## PART 2 选词填空

从方框中选择合适的词，每词只用一次

Read the passage and choose one word for each blank from the box below. Each word may be used only once. There are five extra words.

| | | | |
|---|---|---|---|
| A. adoption | B. advent | C. algorithmic | D. feasibility |
| E. feasible | F. foster | G. interdisciplinary | H. mechanism |
| I. replicate | J. replication | K. scrutiny | L. simulate |
| M. simulated | N. simulation | O. simulations | |

Artificial intelligence is spreading through almost every branch of science. Supporters say it could speed up discovery, much as the microscope and the computer did in earlier centuries. Claims of that kind deserve close __(1)__, since similar promises were made for the telegraph and the internet.

The article argues that these claims rest on a firmer historical foundation. What matters is not the enthusiasm of the moment but the __(2)__ by which a new tool changes how research is done. The __(3)__ of better instruments in the 17th century allowed ideas to travel further and faster; computers later made __(4)__ and modelling possible. Each development altered practice itself, rather than merely speeding up work that was already being done.

Two applications look especially promising. The first examines the existing literature to find connections that researchers have missed; because it draws on several fields at once, the work it encourages is genuinely __(5)__. The second uses robots to form and test hypotheses, and because a machine can be switched on again and again, its experiments are easy to __(6)__ and check.

The main obstacle is not technical. The idea is __(7)__ only if researchers are willing and able to use the tools, and many lack the training. Governments could help by agreeing on common standards, so that different systems can exchange results, and by paying for work that cuts across subjects in order to __(8)__ cooperation between laboratories.

The article notes that the degree of __(9)__ still varies enormously: AI appears in 7.2% of physics and astronomy papers but only 1.4% of those in veterinary science. The author warns, however, about one risk that has nothing to do with speed: __(10)__ bias, in which a machine repeats the unfair assumptions built into its training data.

## PART 3 阅读理解

5 × 2 分

1. What does the article say about earlier technologies such as the electric telegraph and the internet?

A. They were never expected to change society.
B. They were described as solutions to everything, but did not live up to it.
C. They reduced inequality more successfully than AI is likely to.
D. They were rejected by scientists at the time.

2. Why does the author treat the claims about AI more favourably than those earlier claims?

A. AI is funded by governments rather than private firms.
B. New tools have repeatedly led to periods of rapid scientific discovery.
C. AI has already replaced most laboratory work.
D. Scientists now agree on what AI can achieve.

3. The figures 7.2% and 1.4% are used to show that:

A. physics is a more valuable subject than veterinary science.
B. AI is used in science to very different degrees across fields.
C. most scientific papers still contain no AI at all.
D. AI is mainly used for writing papers rather than doing research.

4. According to the article, what is the main thing standing in the way of AI transforming science?

A. The cost of laboratory robots.
B. Whether scientists are willing and able to use the tools.
C. The poor quality of language-based AI systems.
D. A lack of interest from private companies.

5. Why does the article mention Robert Hooke?

A. To show that AI is only another laboratory instrument.

B. To suggest that AI may enlarge human capabilities much as early instruments did.

C. To argue that science has advanced little since the 17th century.

D. To prove that Hooke invented the microscope.

## PART 4 汉译英

将下面段落译成英文

人工智能正在被用于几乎每一个科学领域，但各学科的采用程度差异很大：在 2022 年发表的论文中，物理学和天文学有 7.2% 涉及人工智能，而兽医学只有 1.4%。支持者认为，这类工具能够大幅加快科学发现的速度，正如当年的显微镜、实验室和计算机一样。不过，主要的障碍并非技术性的，而是社会性的：只有当科研人员愿意并且有能力使用这些工具时，这种转变才可能真正发生。

## PART 5 写作

150–200 words

Some people argue that governments should spend more money on artificial intelligence for scientific research, even when the benefits are still uncertain. Others believe that money should go to methods that are already known to work. Discuss both views and give your own opinion. Write 150–200 words.

# ANSWER KEY

参考答案

## VOCABULARY 中文自测答案

scrutiny 仔细审查；详细检查
interdisciplinary 跨学科的
adoption 采用；采纳
algorithmic 算法的
mechanism 机制；运作方式
feasible 可行的；行得通的
foster 促进；培养
simulation 模拟；仿真
replicate 重复；复制
advent 出现；到来

## PART 1 词汇 答案

1. scrutiny
2. mechanism
3. simulation
4. interdisciplinary
5. feasible
6. replicate
7. adoption
8. foster
9. advent
10. algorithmic

## PART 2 选词填空 答案

(1) K scrutiny (2) H mechanism (3) B advent (4) N simulation (5) G interdisciplinary
(6) I replicate (7) E feasible (8) F foster (9) A adoption (10) C algorithmic

## PART 3 阅读理解 答案

1. B 2. B 3. B 4. B 5. B

## PART 4 汉译英 参考译文

Artificial intelligence is now being applied in almost every field of science, but the degree of adoption varies widely: of the papers published in 2022, 7.2% in physics and astronomy involved AI, compared with only 1.4% in veterinary science. Supporters argue that such tools can greatly accelerate the pace of scientific discovery, much as microscopes, laboratories and computers did in their day. The main barrier, however, is not technical but sociological: such a change can take place only if researchers are willing and able to use these tools.

## PART 5 写作 参考范文

Whether public money should be spent on artificial intelligence for science is a question about risk rather than enthusiasm.

Those who favour such spending make a fair point. The article reminds us that microscopes, industrial laboratories and computers each produced bursts of discovery that nobody could have predicted in advance. If governments fund only methods whose value is already proven, they will never pay for the tools that matter most in twenty years. Public funding also matters because private firms have concentrated almost all their effort on language-based systems, leaving other approaches short of support.

The opposing view is also reasonable. Money spent on an uncertain technology is money not spent on teaching, equipment or research that already works. The article itself notes that the main barrier is social rather than technical: if scientists lack the skills to use these tools, extra funding will not help.

In my opinion, the two positions can be combined. Governments should fund AI research alongside proven methods, and should attach clear conditions: shared data standards, training for researchers, and public results. Funding is worthwhile when it is designed so that the knowledge gained remains useful even if the original hope fails.